

8 | Ground Truth Data

For a prediction model to be successful, it needs to have the proper training. This training-data needs to be as close to 100% certain as possible. However for the topic of gentrification, it is not as straight forward, since there is no clear definition of gentrification in the literature. Previous studies like Drew (2018) have compiled different approaches on to how to quantify gentrification, but most of these studies are hindered by limited data. The ML model developed in this project uses only data from The Danish Neighborhood Atlas (DNA). Since there are relevant 20 parameters to choose from, the task then becomes how to weigh each layer.

8.1 Space Time Analysis

This section will explain how the raw data from DNA is processed into Time Series Clusters, so that it can be reclassified to show which clusters show signs of gentrification.

Before making a ML model it is necessary to get an understanding of the chosen data from DNA and its patterns. With a Space Time Analysis (STA) the changes in data through time tied to a specific location is analyzed (ESRI, n.d.-b). The tool used to analyze these trends are Time Series Clustering in ArcGIS Pro. This tool gathers data defined by how the data for each location have been evolving over time. This shows which areas have undergone a similar change. Some neighborhoods might have been through a steady rise following the municipal or national mean, while others have seen a drastic surge at a specific period in time. The main aim of the analysis is to find areas that have been gentrified and can be used as ground truth data (GTD) in the machine learning model.

8.2 Scope of Analysis

In the limited literature on gentrification in Denmark, especially Inner Vesterbro stands out as gentrified but also Nørrebro, however to a slightly lesser extent. Therefore a scope is needed to capture the development of these neighborhoods. The area for the Space Time Analysis (STA) was the entire city of Copenhagen (Frederiksberg and Copenhagen municipal borders included). The map on figure 8.1 shows the borders for both the

different suburbs in Copenhagen and Frederiksberg municipality. Choosing the whole of Frederiksberg and Copenhagen made it possible to also investigate other, smaller gentrified neighborhoods inside Copenhagen, that was unknown.

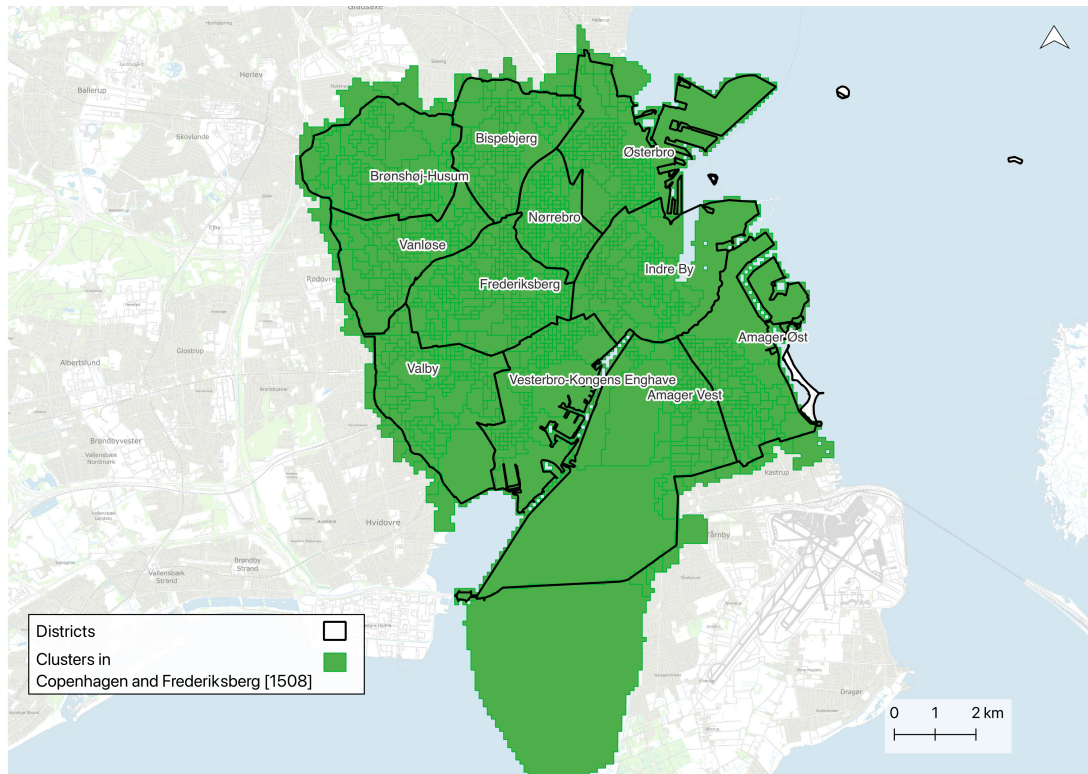


Figure 8.1: Neighborhoods in Copenhagen and Frederiksberg (1508 individual clusters)

Originally the Space Time Analysis was intended to explore the whole study area of the Finger Plan¹. However when using the STA-tools on the whole area it was quickly realized that this approach made it impossible to see trends in the data with the level of detail needed. As a result the scope was narrowed down to only the neighborhoods in Copenhagen. This decision is based on the literature, that points out the Copenhagen area to be most likely to be exposed to gentrification, as described in section 3.2. This allowed us to see trends without the risk of overfitting the data. When developing and testing the final ML model the outline of the Finger Plan will be used. This data will be used as test data and will be explained further in chapter 9.

¹The Finger Plan is a plan document from 1947 and details the planning for the greater Copenhagen area, including northern and southern Sjælland as well as Roskilde. The plan spans 34 municipalities (Erhvervsstyrelsen, 2019).

8.3 Application of Tools

The whole workflow of the GTD is done in Python scripts and can be found in the GitHub repository. Figure 8.2 shows which functions are applied to the raw DNA data. The coding is split up into four distinct scripts. The first script to process the data is a python-toolbox for ArcGIS Pro and all other functions and scripts are made in open-data platforms, such as GeoPandas, QGIS and Rasterio.

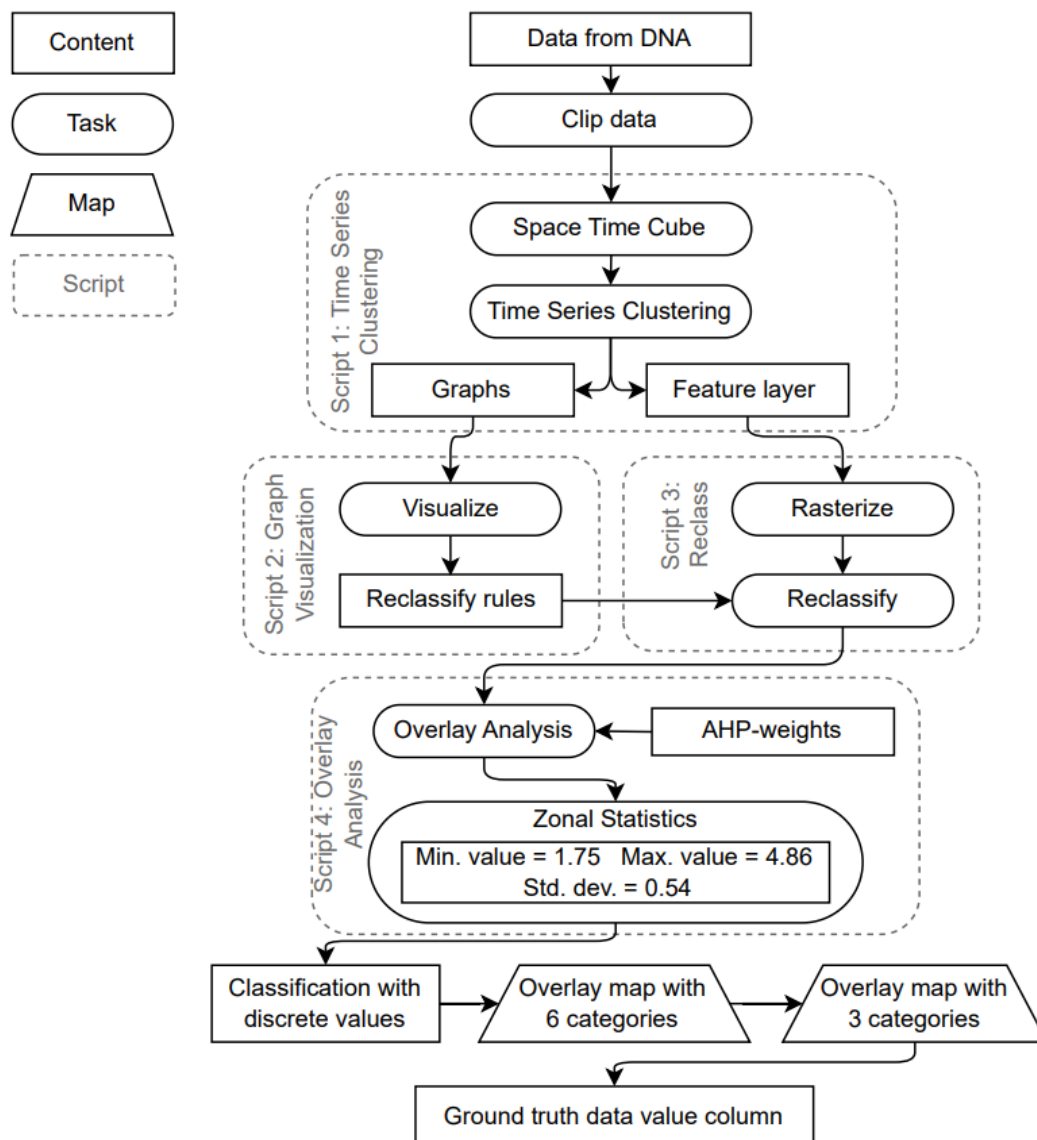


Figure 8.2: Workflow of processing the ground truth data

Before the different spatial tools are applied to the data, a rough sorting is needed. As mentioned earlier in chapter 7, there are few neighborhoods that have seen a large increase in population from 1990 to 2020 because of new construction e.g. Ørestaden

and Carlsbergbyen. As a result, these neighborhoods are too irregular when determining how they fit into the scope of gentrification. Also large parks and ponds are removed to further clean the data. This means that all clusters over 50ha are being removed from the training data as well as from the test data. Neighborhoods larger than 50ha will at some point from 1990 to 2020 have been too sparsely populated and potentially cause problems for the ML model. A map of the final clusters can be seen on figure 8.3

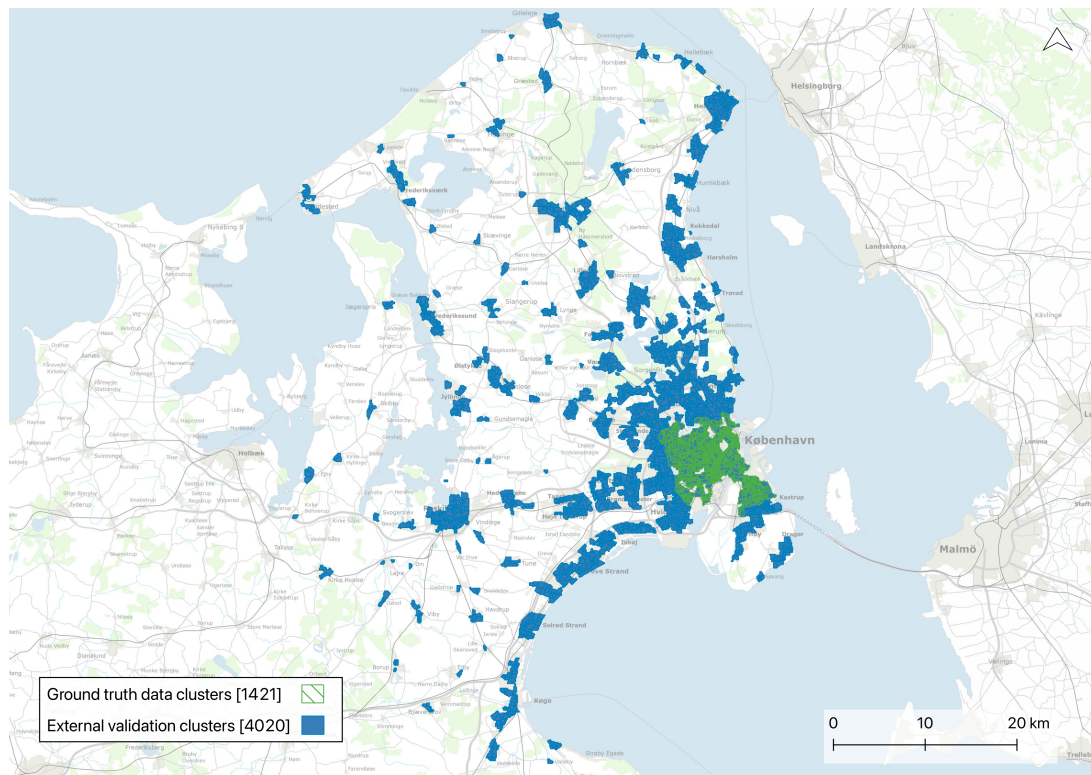


Figure 8.3: Remaining clusters after sorting out clusters over 50ha

This leaves 1,421 clusters for ground truth data and 4,020 clusters as testing data (including the 1,421 GTD clusters). In the next section the workflow of compiling the GTD is further elaborated.

8.3.1 Time Series Clustering

In preparation the data has been turned from CSV files into a compatible format and then into 'space time cubes'. A space time cube is the data structure used as the foundation for the different tools in the 'space time toolbox' (ESRI, n.d.-a). The tool 'time series clustering' is applied and shows emerging patterns in each neighborhood. The tool analyzes each neighborhood at each timestep/year and then connects similar patterns

together (ESRI, n.d.-b). This makes it possible to select the number of clusters that the data is grouped into. Here the number was set to six individual clusters.

A lower number of clusters showed a lower level of detail in the patterns making them too general to reveal the relevant information. Contrary, a larger number of clusters made it difficult, because it showed too much detail and created noise in the data. Upwards of 20 clusters were tested during this phase but the data was ultimately too noisy to see patterns among the different neighborhoods. Time series clustering then groups clusters that have the same development over time together and generates a shape layer containing six different values corresponding to these different developments.

Figure 8.4 is showing the result of the layer 'Unemployment', which clusters each neighborhood, as well as graphs showing how the clusters changes through time. For unemployment it is argued that a large amount of unemployed residents, that is decreasing quickly over time, is more evident of gentrification than a low number of unemployment with little decrease. This can be seen on figure 8.4a, where the light blue line (Cluster 6) is considered a bigger contributor to gentrification than the purple line (Cluster 4). This process has been done to all the 20 different variables from DNA, where an argumentation for each categorization can be seen in appendix A.

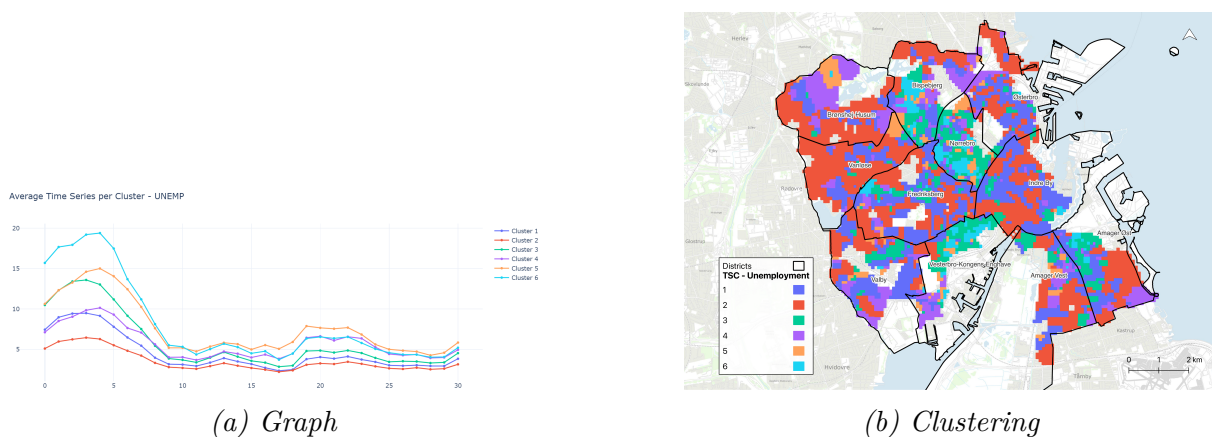


Figure 8.4: Time Series Clustering showing the changes in unemployment rate

8.3.2 Graph Visualization

A small script was written to get a better overview of the output graphs. The graph-files are all converted to figures and can be seen in appendix A.

8.3.3 Reclassification

After each time series cluster had been examined, each individual graph was evaluated and reclassified by how each graph of the clusters matched the understanding of gentrification from the literature review. Each graph was also compared to both municipal and national means from the Danish Neighborhood Atlas to see the changes in a broader perspective.

This led to the conclusion that a big change in e.g. unemployment rate meant that a new demographic composition had occurred and therefore gentrification presumably has taken place at some stage. This can be seen on figure 8.4b which shows unemployment rate divided into six classes.

Meanwhile the areas with a smaller decrease in unemployment rate shows that the demographic composition was stable through this time period from 1990 to 2020. This meant in practice that the brown graph on figure 8.4 was attributed the new value 1 and the yellow graph got value 6. This process was repeated for each dataset linking the changes in the data to how gentrification is measured and seen in practice². See appendix A for further insight.

8.3.4 Overlay Analysis

The aim of the analysis is to show areas that have been subject to gentrification and therefore a map is needed that encapsulates all of these different developments. With a weighted overlay analysis each dataset gets taken into account and weighed by how well it can be used as a measurement of gentrification (ESRI, n.d.-c).

As explained in the literature review in chapter 3, the concept of gentrification is a complex phenomenon with several different definitions and criteria. In the case of Vidal (2019), Drew (2018) and Reades et al. (2019) only a few variables are used as measurements (Square-meter price, disposable income, higher education and unemployment), whereas Institute on Metropolitan Opportunity (2016), Yoo (2023) and Knorr (2019) supplements with several other variables (family composition, age, ethnicity, migration rate etc.).

Because of the different understanding on which data to be used and how, it is decided that the overlay analysis should consist of the DNA data and be weighted on the basis of an 'analytic hierarchy process' (AHP). AHP is a multi-criteria decision method used by

²The reclassification of gentrification-values can be found in the GitHub repository under data/interim/reclass_tsc.xlsx